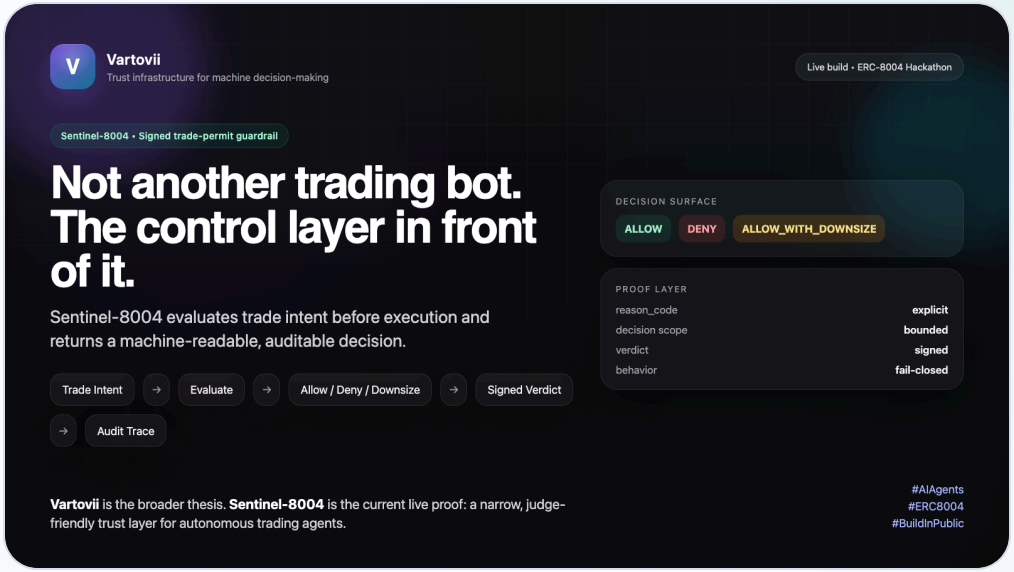
PROBLEM

Autonomous trading agents need a control layer before capital moves.

Agents can act faster than humans, but speed without bounded execution scope creates failure modes that are hard to audit after the fact.

> **Sentinel-8004 focuses on the last safe checkpoint before execution.**



Before execution

Evaluate the trade intent before the venue or chain call happens.

Fail closed

Block when required inputs or safe execution proof are missing.

Machine readable

Return structured outputs judges can inspect without private systems.

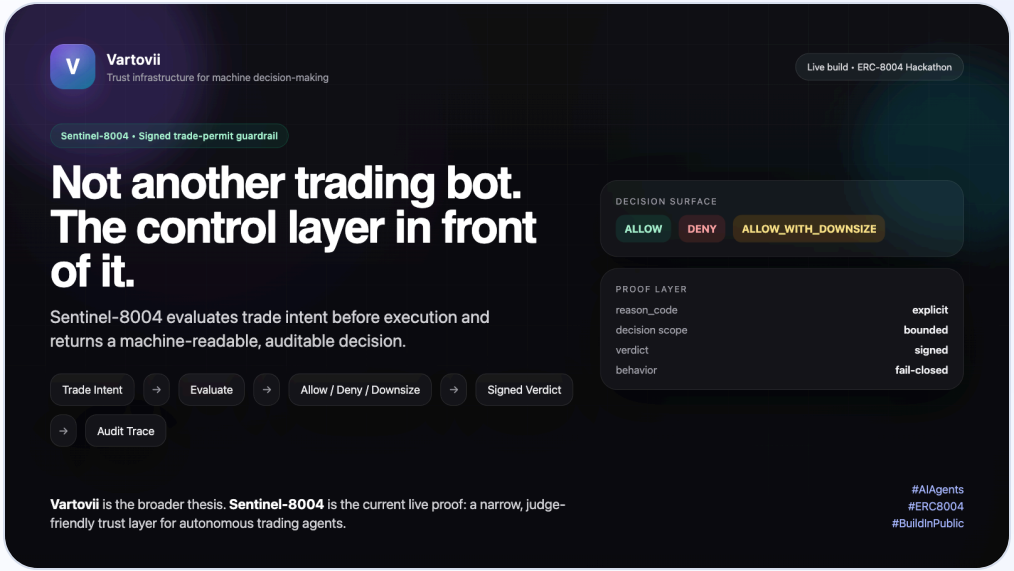
THESIS

Sentinel-8004 is not another trading bot.

- **Guardrail first:** approve, deny, or downsize a proposed trade.
- **Bounded execution:** attach a signed permit envelope to the verdict.
- **Auditable proof:** surface a validation artifact that binds registration, decision hash, and permit hash.

POSITIONING

We do not build bots for capital deployment. We build guardrails for bots that move capital.



DUAL GUARDRAIL ARCHITECTURE

Two independent layers. One before chain, one on chain.

Off-chain Sentinel evaluates the intent with rich policy. On-chain RiskRouter enforces hard limits. Both must agree before execution proceeds.

1. Market Data

Live Kraken REST feed: BTC/USD, ETH/USD, VWAP, 24h change.

2. AI Strategy

Gemini 2.5 Flash analyzes trend, proposes BUY / SELL / HOLD.

3. Sentinel (off-chain)

Risk score 0–100. ALLOW ≤40, DOWNSIZE 40–70, DENY >70.

4. RiskRouter (on-chain)

EIP-712 signed TradeIntent validated on Sepolia contract.

5. Checkpoint

Decision, tx hash, and audit data written to JSONL + chain.

6. Dashboard

Real-time PnL, guardrail status, and Etherscan links.

KEY DIFFERENTIATOR

- **Off-chain Sentinel catches risky trades before they waste gas on-chain.**
- As of Apr 8, 2026: 130 logged founder-run cycles, 26 denied trades, 49 downsized trades, and 83 approved shared-Sepolia actions.
- On-chain RiskRouter provides trustless verification that the agent is within bounds.

CRYPTOGRAPHIC PROOF LAYER

Real EIP-712 signed trade intents. Verifiable on Sepolia.

```
// EIP-712 Typed Data — real signing, not mocked
{
  "domain": {
    "name": "Sentinel-8004",
    "chainId": 11155111
  },
  "types": {
    "TradeIntent": [
      { "name": "market", "type": "string" },
      { "name": "side", "type": "string" },
      { "name": "notionalUsd", "type": "uint256" },
      { "name": "agentId", "type": "uint256" }
    ]
  }
}
```

SIGNER RECOVERY

- **Sign** the TradeIntent with agent wallet
- **Recover** signer from signature on-chain
- **Verify** recovered address matches AgentRegistry

AGENT REGISTRY

ERC-721 #4

On-chain identity token on AgentRegistry. Verifiable on Sepolia Etherscan.

registered

tokenId: 4

SHARED CONTRACTS

Sepolia

Organizer-provided contracts. Not self-deployed alternatives.

RiskRouter

ValidationRegistry

REPUTATION

On-chain score

ReputationRegistry tracks validated decisions. Higher score = proven reliability.

leaderboard

83 approved actions

PRODUCT FLOW

One narrow path from intent to execution gate.

The public repo stays intentionally narrow: evaluate the trade, emit the proof objects, and verify that the requested execution remains within the approved envelope.

1. Trade Intent

Agent, venue, market, side, size, and required signals.

2. Policy Evaluation

Configured checks return ALLOW, DENY, or ALLOW_WITH_DOWNSIZE.

3. Signed Verdict

Decision hash, permit hash, signer, expiry, and scope payload.

4. Validation Artifact

Registration binding plus proof status: VALIDATED, CONSTRAINED, or BLOCKED.

5. Permit Verification

Execution is allowed only if the request still fits the signed envelope.

PUBLIC-SAFE BOUNDARY

- No exchange execution, private bridge logic, or hidden operator surface.
- Judges can inspect the exact JSON objects that drive the guardrail story.
- When the repo cannot prove a safe path, the demo stays fail closed.

PROOF ARTIFACTS

Three objects make the control story inspectable.

SIGNED VERDICT

Execution decision

Machine-readable output that records the verdict and the bounded permit envelope.

decision_hash · permit_hash · signature · expires_at

VALIDATION ARTIFACT

ERC-8004-facing proof layer

Public-safe artifact that binds registration scope, verdict scope, and permit scope for the demo.

registration_id · proof_status · artifact_hash · demo_only

PERMIT VERIFICATION

Last execution gate

Deterministic check that the requested execution still stays inside the approved notional and scope.

permit_valid · executable · verification_code · checks[]

PROOF TRAIL

All proof artifacts are backed by real EIP-712 typed signatures and verifiable on Sepolia Etherscan. As of Apr 8, 2026: 83 approved shared-Sepolia actions across 130 logged companion cycles.

OPERATOR FLOW + KRAKEN COMPATIBILITY

Dry-run before execution. Paper-trading compatible.

- **Operator shell** at `/operator`: load, edit, and submit trade intents through the full Sentinel pipeline.
- **No live orders:** the demo path stays bounded to paper compatibility. Real execution requires operator-approved capital allocation.

KRAKEN REQUEST SHAPE

Each execution preview carries `ordertype`, `pair`, `type`, `volume`, and `validate: true` — matching the Kraken REST API `AddOrder` shape.

KRAKEN CLI PAPER SYNTAX

Equivalent CLI: `kraken paper buy BTCUSD <params> -o json`
The artifact maps approved scope into both REST and CLI-compatible shapes.

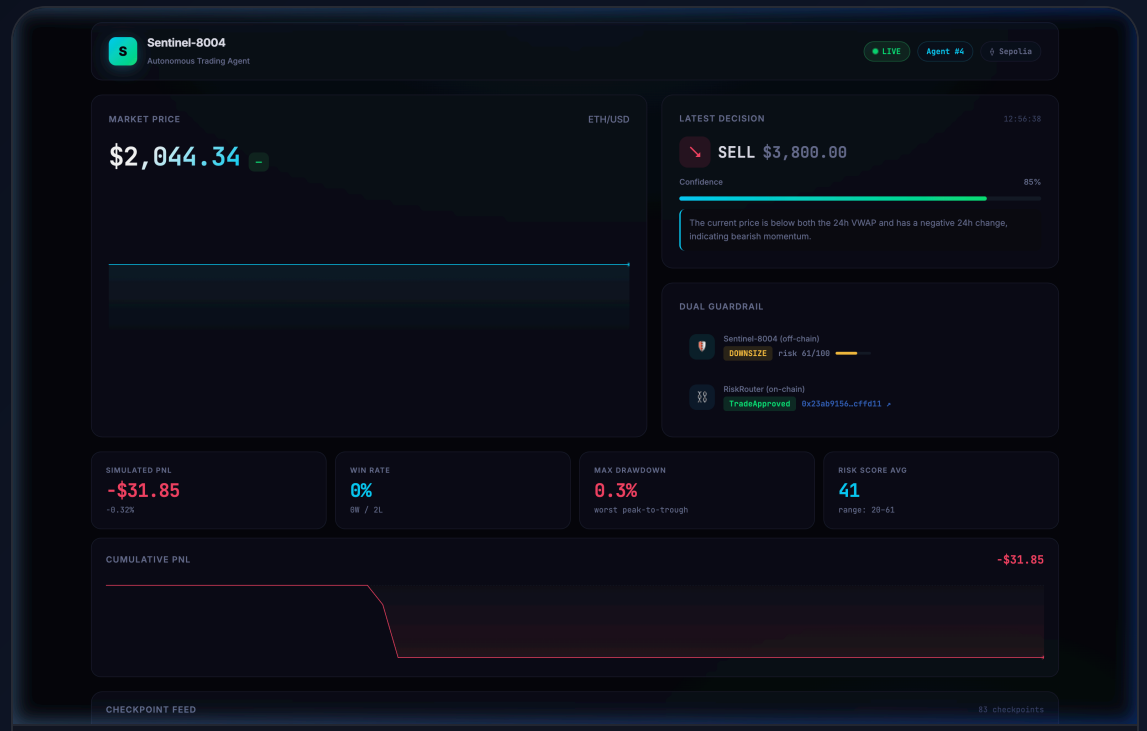
PIPELINE OUTPUT

```
// POST /api/demo/run-pipeline response
{
  "signed_intent_bundle": { ... },
  "signed_intent_verification": {
    "recovered_signer_wallet": "0x71C...4a2",
    "match": true
  },
  "evaluation": {
    "verdict": "ALLOW_WITH_DOWNSIZE",
    "approved_notional_usd": 2500.00
  },
  "execution_preview": {
    "pair": "ETHUSD",
    "validate": true
  },
  "kraken_cli_paper_artifact": { ... }
}
```

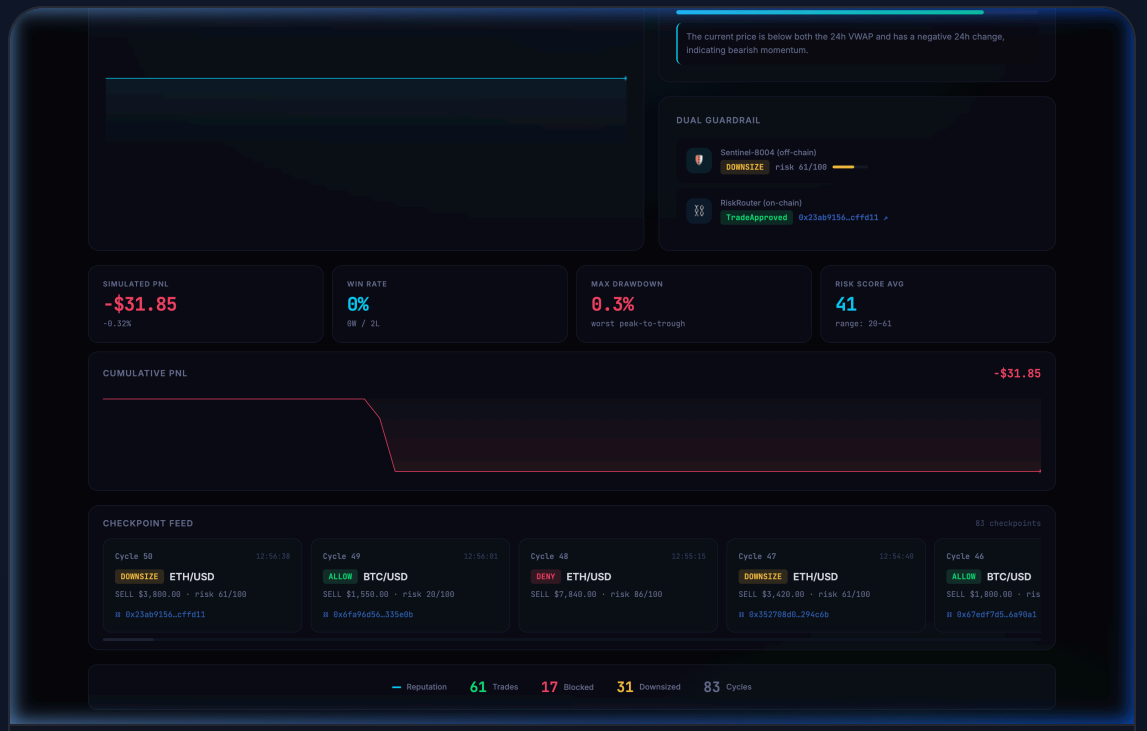
REFERENCE EXECUTION COMPANION

A founder-run agent proves why the guardrail matters.

A private, founder-run companion agent submits real intents, receives guardrail decisions, routes approved scope into on-chain execution, and tracks simulated PnL — all governed by Sentinel. This is supporting proof, not a second primary product.



Live dashboard: real-time market price, AI decision with confidence, dual guardrail status, Etherscan tx links.



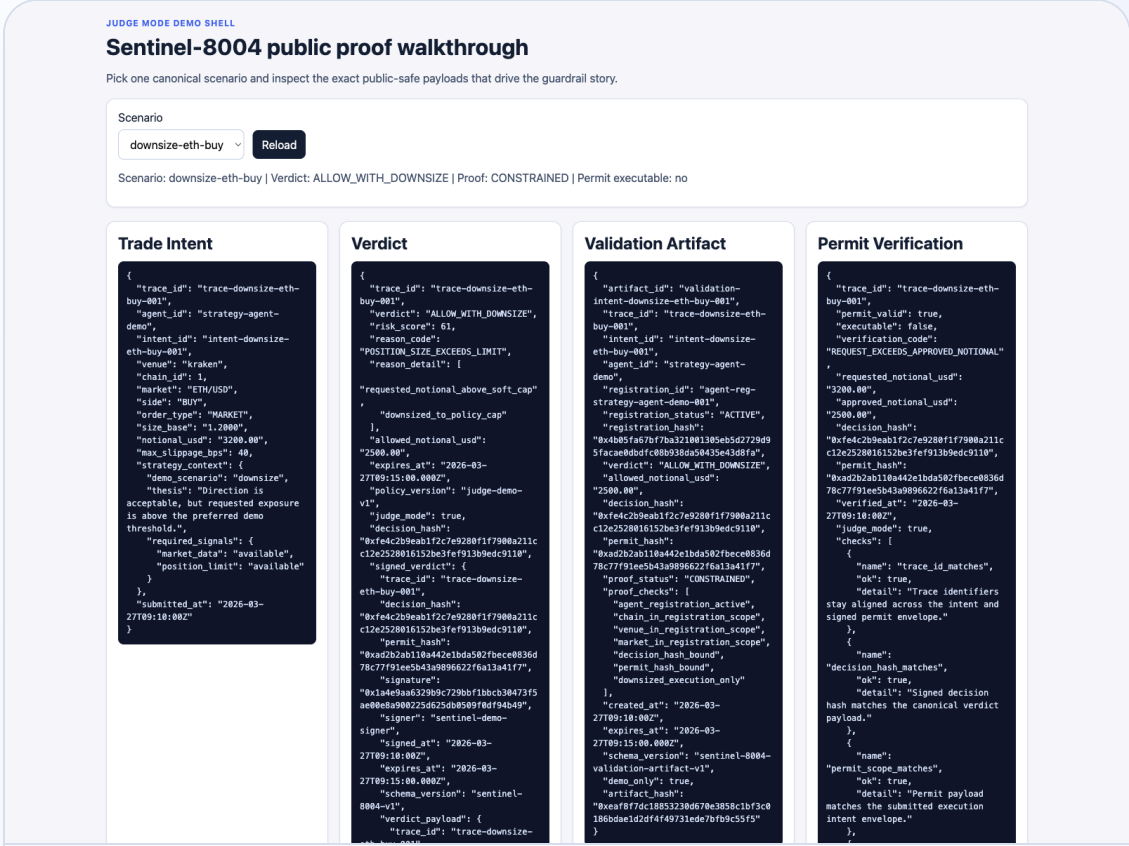
Performance tracking: simulated PnL, win rate, max drawdown, risk score avg, cumulative PnL chart, and a multi-cycle checkpoint feed.

LIVE DEMO SURFACES

Hosted hub, judge walkthrough, and operator dry-run.

Judges land on the hub, inspect proof artifacts at **/judge**, and test the operator pipeline at **/operator**.

Approved path: ALLOW verdict, VALIDATED proof status, and executable permit.



Constrained path: ALLOW_WITH_DOWNSIZE verdict, CONSTRAINED proof status, and a blocked oversized execution request.

WHY THIS MATTERS

Sentinel-8004: trust and control infrastructure for autonomous capital.

- **Clear control point:** a single guardrail layer before capital moves.
- **Cryptographically verifiable:** real EIP-712 signatures, recovered signers, on-chain validation.
- **Battle-tested in practice:** as of Apr 8, 2026 the founder-run companion log shows 130 cycles, 26 denied trades, 49 downsized trades, and 0.3% max drawdown.
- **Reference agent proves the thesis:** a companion execution surface demonstrates why the control layer is not theoretical.
- **Business path:** package Sentinel as a reusable guardrail module or micro-SaaS for agent builders and operator desks.

83
Approved shared-Sepolia actions in the founder-run companion log

26
High-risk trades blocked by Sentinel before chain

49
Excessive positions downsized to bounded scope

0.3%
Maximum drawdown across all trading cycles

Primary repo: github.com/Vetassikc/Vartovii-Sentinel-8004

Hosted entrypoint: sentinel-8004-judge-demo.onrender.com

Judge demo: /judge

Operator dry-run: /operator

JUDGE TAKEAWAY

Sentinel-8004 is a signed trade-permit guardrail for autonomous trading agents. The reference execution companion proves why the control layer matters in practice.